IGCSE Chemistry CIE

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8. The Periodic Table

CONTENTS

8.1 The Periodic Table & Trends

8.1.1 The Periodic Table

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The Periodic Table

- There are over 100 chemical elements which have been isolated and identified
  - Each element has one proton **more** than the element preceding it
  - This is done so that elements end up in columns with other elements which have similar properties
  - Elements are arranged on the periodic table in order of **increasing atomic number**
  - The table is arranged in vertical columns called **groups** and in rows called **periods**

- **Period:** These are the horizontal rows that show the number of shells of electrons an atom has and are numbered from 1 – 7
  - E.g. elements in period 2 have two electron shells, elements in period 3 have three electron shells

- **Group:** These are the vertical columns that show how many outer electrons (also known as valency electrons) each atom has and are numbered from I – VII, with a final group called Group 0 (instead of Group VIII)
  - E.g. Group IV elements have atoms with 4 electrons in the outermost shell, Group VI elements have atoms with 6 electrons in the outermost shell and so on

- The group number can help determine the charge that metal and non-metal ions form
- For metals, the group number corresponds to the number of electrons it will lose to achieve a full outer shell and the charge of the metal ion
  - E.g. sodium is in **Group I,** it will lose 1 electron and form an ion with a **1+ charge**
  - Magnesium is in **Group** II, it will lose 2 electrons and form an ion with a **2+ charge**

- For non-metals in Group VII and VI, they will gain 1 and 2 electrons respectively to gain a full outer shell
  - E.g. non-metals in **Group VII** gain 1 electron to form ions with a **1– charge**
  - Non-metals in **Group VI** gain 2 electrons to form ions with a **2– charge**

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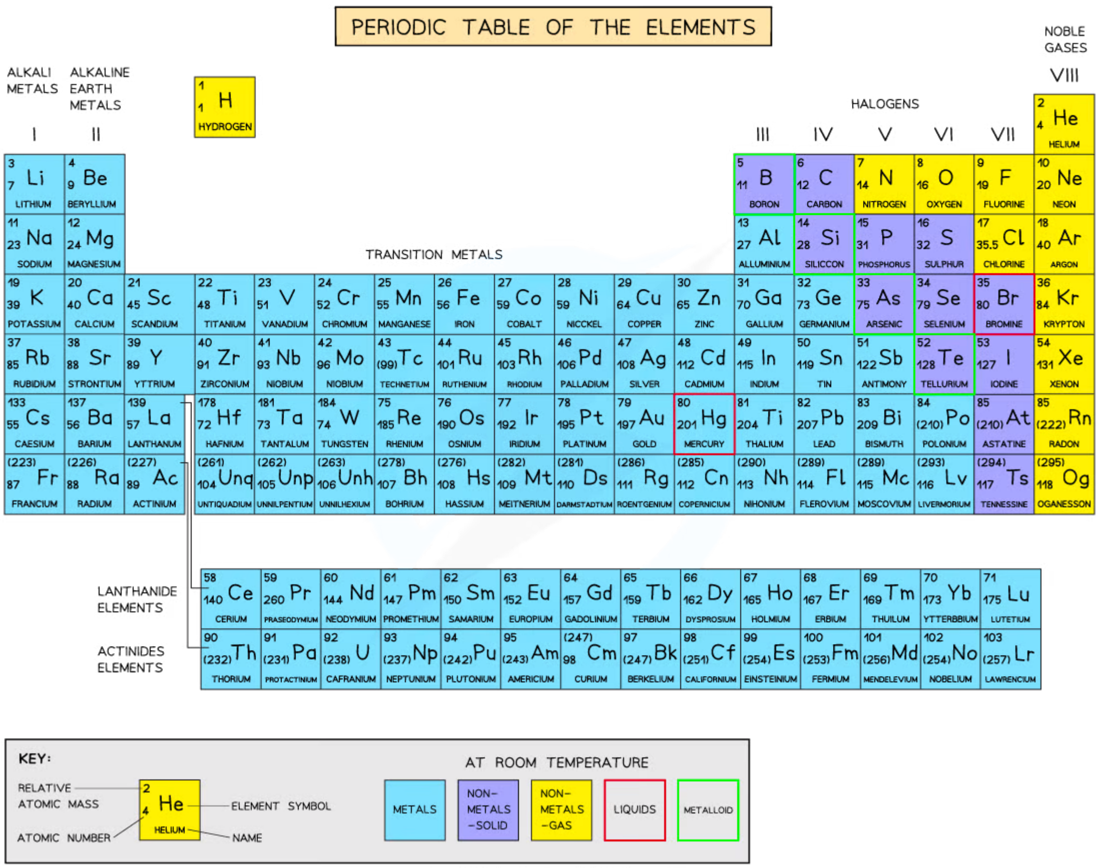

All elements are arranged in the order of increasing atomic number from left to right

Valency

- Valency (or combining power) tells you how many bonds an atom can make with another atom or how many electrons its atoms lose, gain or share, to form a compound
  - E.g. carbon has a valancy of 4 as it is in Group IV so a single carbon atom can share 4 electrons to make 4 single bonds or 2 double bonds
- The following valencies apply to elements in each group:

GROUP	VALENCY
I	1
II	2
III	3
IV	4
V	3
VI	2
VII	1
VIII	0

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Exam Tip

An easier way of remembering which number is the mass number and which is the atomic is:

Mass Number = The **mass**ive number i.e the larger of the two numbers.

The atomic number must be the smaller number.

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The Metallic Character of Elements

- The metallic character of the elements **decreases** as you move across a Period on the Periodic Table, from **left to right**, and it **increases** as you move down a Group
- This trend occurs due to atoms more **readily accepting** electrons to fill their valence shells rather than losing them to have the previous, already full, electron shell as their outer shell
- Metals occur on the **left-hand** side of the Periodic Table and non-metals on the **right-hand** side
- Between the metals and the non-metals lie the elements which display some properties of **both**
- These elements are referred to as metalloids or semi-metals

Properties of metals and non-metals

PROPERTY	METALS	NON-METALS
ELECTRON ARRANGEMENT	1 – 3 (MORE IN PERIODS 5 & 6) OUTER SHELL ELECTRONS	4 – 7 ELECTRONS IN THE OUTER SHELL
BONDING	METALLIC DUE TO LOSS OF OUTER SHELL ELECTRONS	COVALENT BY SHARING OF OUTER SHELL ELECTRONS
ELECTRICAL CONDUCTIVITY	GOOD CONDUCTORS OF ELECTRICITY	POOR CONDUCTORS OF ELECTRICITY
TYPE OF OXIDE	BASIC OXIDES (A FEW ARE AMPHOTERIC)	ACIDIC OXIDES (SOME ARE NEUTRAL)
REACTION WITH ACIDS	MANY REACT WITH ACIDS	DO NOT REACT WITH ACIDS
PHYSICAL CHARACTERISTICS	MALLEABLE, CAN BE BENT AND SHAPED HIGH MELTING AND BOILING POINT	FLAKY, BRITTLE LOW MELTING AND BOILING POINT

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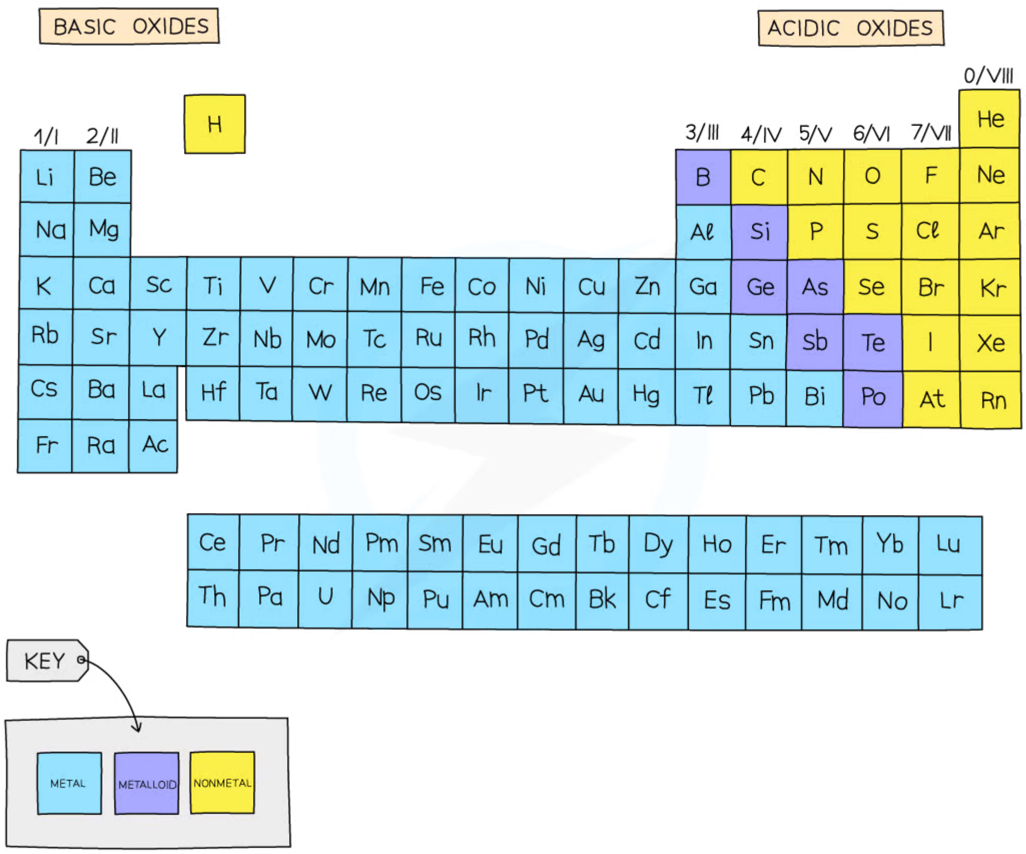

A zig-zag line in this diagram separates the metals on the left, from the non-metals on the right

Periodic Trends & Electronic Configuration

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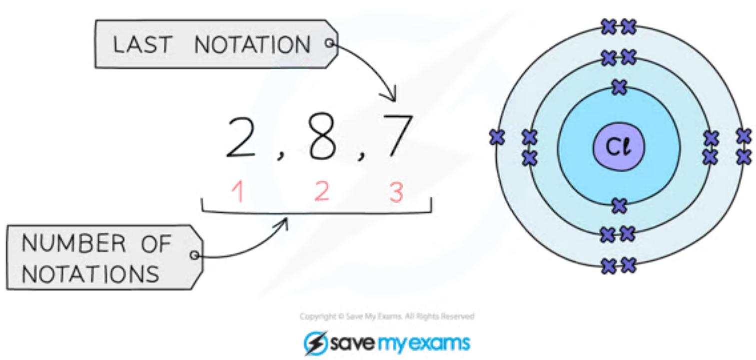
- The electronic configuration is the arrangement of electrons into shells for an atom (e.g: the electronic configuration of carbon is 2,4)
- There is a link between the electronic configuration of the elements and their position on the Periodic Table
- The number of notations in the electronic configuration will show the number of occupied shells of electrons the atom has, showing the **period**
- The last notation shows the number of outer electrons the atom has, showing the **group** number

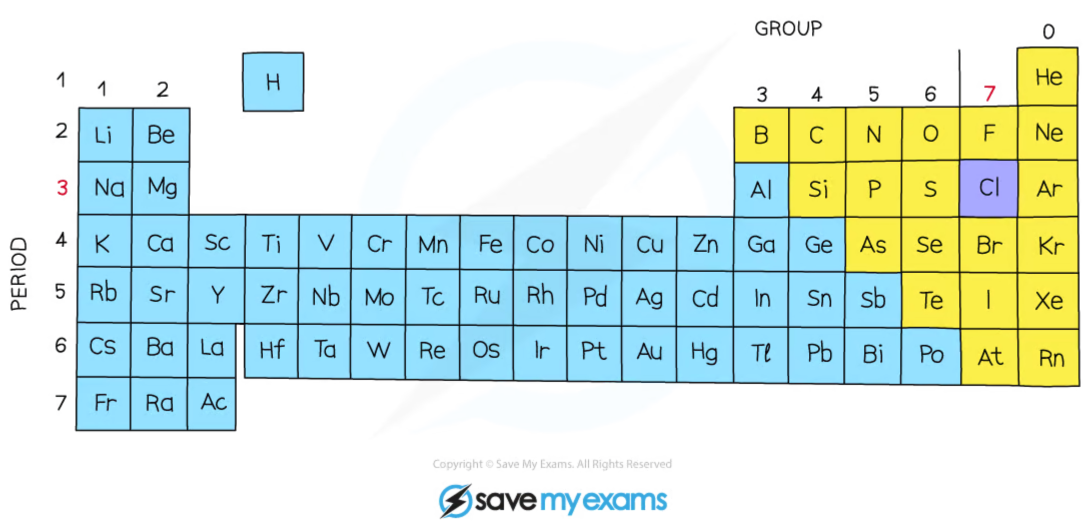
Example: Electronic configuration of **chlorine**:



The electronic configuration of chlorine as it should be written

Period: The red numbers at the bottom show the number of notations which is 3, showing that a chlorine atom has 3 shells of electrons.

Group: The final notation, which is 7 in the example, shows that a chlorine atom has 7 outer electrons and is in Group VII



The position of chlorine on the Periodic Table

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- Elements in the same group in the Periodic Table have **similar chemical properties**
- When atoms collide and react, it is the **outermost electrons** that interact
- The similarity in their chemical properties stems from having the **same number** of electrons in their outer shell
- For example, both lithium and sodium are in Group I and can react with elements in Group VII to form an ionic compound (charges of Group I ions are 1+, charges of Group VII ions are 1–) by reacting in a similar manner and each donating one electron to the Group VII element
- As you look down a group, a full shell of electrons is added to each subsequent element
  - Lithium's electronic configuration: 2,1
  - Sodium's electronic configuration: 2,8,1
  - Potassium's electronic configuration: 2,8,8,1

> **Exam Tip**
> Electronic configurations can be shown with the numbers separated by commas or by full stops. In this course commas are used, but you will often see full stops used elsewhere. Both are accepted.

Predicting Properties

- Because there are **patterns** in the way the elements are arranged on the Periodic Table, there are also **patterns** and **trends** in the chemical behaviour of the elements and their physical properties
- These trends in properties occur down groups and across the periods of the Periodic Table
- As a result, we can use the Periodic Table to predict properties such as:
  - boiling point
  - melting point
  - density
  - reactivity
- Some common properties / trends in properties include:
  - Group I elements react very quickly with water
  - Noble gases are unreactive
  - Transition elements are denser than Group I elements
  - Reactivity decreases going down Group VII
  - Melting point decreases going down Group I
- In this way the Periodic Table can be used to **predict** how a particular element will behave

Identifying Trends

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- Using given information about elements, we can identify trends in properties
- An example of when this might be used is to determine the trend in reactivity of Group I metals
- The table below shows the reactions of the first three elements in Group I with water

Observations of Lithium, Sodium, and Potassium with Water

Element	Reaction
Lithium	$2Li(s) + 2H_2O(l) \longrightarrow 2LiOH(aq) + H_2(g)$ ∘ Reaction slower than with sodium ∘ Bubbles of hydrogen gas ∘ Lithium's melting point is higher and heat isn't produced so quickly, so the lithium doesn't melt
Sodium	$2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g)$ ∘ Bubbles of hydrogen gas ∘ Melts into a shiny ball that dashes around the surface ∘ Floats on water because it is less dense ∘ Melts because sodium has a low melting point and a lot of heat is made in the reaction ∘ Hydrogen is evolved which causes the ball of sodium to move around the surface of the water ∘ White trail of sodium hydroxide produced which dissolves in the water producing a highly alkaline solution
Potassium	$2K(s) + 2H_2O(l) \longrightarrow 2KOH(aq) + H_2(g)$ ∘ Reacts more violently than sodium ∘ Bubbles of gas ∘ Melts into a shiny ball that dashes around the surface ∘ Enough heat produced so burns with a lilac–coloured flame

- The observations show that reactivity of the Group I metals increases as you go down the group
- Using this information we can predict the trend going further down Group I for the elements rubidium, caesium and francium
- As the reactivity of alkali metals increases down the group, rubidium, caesium and francium will react more vigorously with air and water than lithium, sodium and potassium
- Lithium will be the **least** reactive metal in the group at the top, and francium will be the **most** reactive at the bottom
- Francium is rare and radioactive so is difficult to confirm predictions

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Table to Show the Predicted Reaction of other Group I Elements with Water

ELEMENT	REACTION
RUBIDIUM	– EXPLODES WITH SPARKS – RUBIDIUM HYDROXIDE PRODUCED.
CAESIUM	– VIOLENT EXPLOSION DUE TO RAPID PRODUCTION OF HEAT AND HYDROGEN – CAESIUM HYDROXIDE PRODUCED.
FRANCIUM	– TOO REACTIVE TO PREDICT

Exam Tip

For the extended course you may be asked to identify other trends in chemical or physical properties of Group I metals, given appropriate data.

Firstly, ensure that the metals and associated data are written in either descending or ascending order according the their position in the Group. Then look for general patterns in the data.

8.2 Group Properties & Trends

8.2.1 Group I Properties

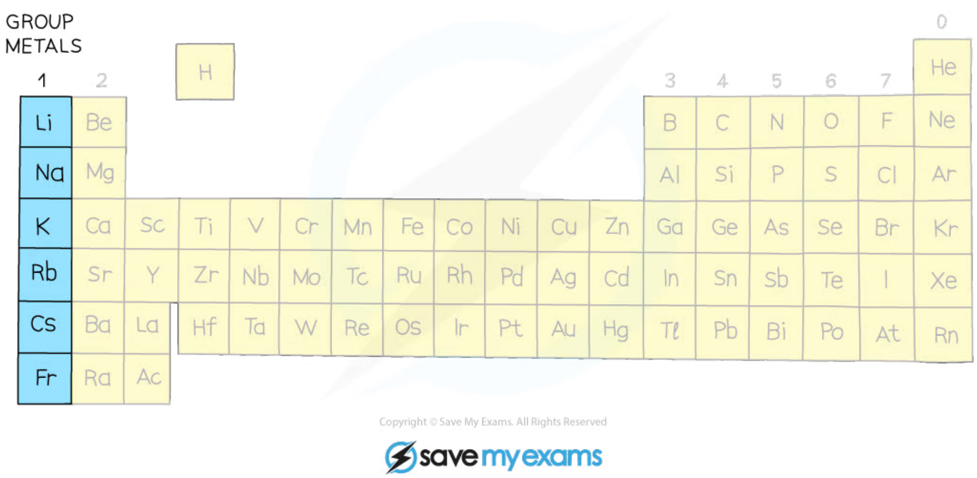
Group I Properties & Trends: Basics

The Group I metals

- The Group I metals are also called the alkali metals as they form **alkaline solutions** with high pH values when reacted with water
- Group I metals are lithium, sodium, potassium, rubidium, caesium and francium
- They all contain just **one electron** in their outer shell

Physical properties of the Group I metals

- The Group I metals:
  - Are soft and easy to cut, getting even **softer** and **denser** as you move down the Group (sodium and potassium do not follow the trend in density)
  - Have **shiny** silvery surfaces when freshly cut
  - Conduct **heat** and **electricity**
  - They all have **low** melting points and **low** densities compared to other metals, and the melting point **decreases** as you move down the Group; some would melt on a hot day



The alkali metals lie on the far left-hand side of the Periodic Table

Chemical properties of the Group I metals

- They react readily with oxygen and water vapour in air so they are stored under **oil** to stop them from reacting
- Group I metals will react similarly with water, reacting vigorously to produce an **alkaline** metal hydroxide solution and **hydrogen** gas
- The Group I metals get more reactive as you look down the group, so only the first three metals are allowed in schools for demonstrations

Reactions of the Group I metals and water

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Element	Reaction
Lithium	$2Li(s) + 2H_2O(l) \longrightarrow 2LiOH(aq) + H_2(g)$ ∘ Reaction slower than with sodium ∘ Bubbles of hydrogen gas ∘ Lithium's melting point is higher and heat isn't produced so quickly, so the lithium doesn't melt
Sodium	$2Na(s) + 2H_2O(l) \longrightarrow 2NaOH(aq) + H_2(g)$ ∘ Bubbles of hydrogen gas ∘ Melts into a shiny ball that dashes around the surface ∘ Floats on water because it is less dense ∘ Melts because sodium has a low melting point and a lot of heat is made in the reaction ∘ Hydrogen is evolved which causes the ball of sodium to move around the surface of the water ∘ White trail of sodium hydroxide produced which dissolves in the water producing a highly alkaline solution
Potassium	$2K(s) + 2H_2O(l) \longrightarrow 2KOH(aq) + H_2(g)$ ∘ Reacts more violently than sodium ∘ Bubbles of gas ∘ Melts into a shiny ball that dashes around the surface ∘ Enough heat produced so burns with a lilac–coloured flame

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Predicting the Properties of Group I Elements

- Knowing the reactions of elements at the top of the group allows you to predict the properties of other elements further down Group I

Properties of other Alkali Metals (Rubidium, Caesium and Francium)

- As the reactivity of alkali metals increases down the group, rubidium, caesium and francium will react more vigorously with air and water than lithium, sodium and potassium
- Lithium will be the **least** reactive metal in the group at the top, and francium will be the **most** reactive at the bottom
- Francium is rare and radioactive so is difficult to confirm predictions
- For example the reactions with water can be predicted:

Predicting the Reaction with Water

ELEMENT	REACTION
RUBIDIUM	– EXPLODES WITH SPARKS – RUBIDIUM HYDROXIDE PRODUCED.
CAESIUM	– VIOLENT EXPLOSION DUE TO RAPID PRODUCTION OF HEAT AND HYDROGEN – CAESIUM HYDROXIDE PRODUCED.
FRANCIUM	– TOO REACTIVE TO PREDICT

- You can also look at other properties such as boiling point, melting point and density of Group I elements and use them to predict whether the other properties are likely to be larger or smaller going down the group

8.2.2 Group VII Properties

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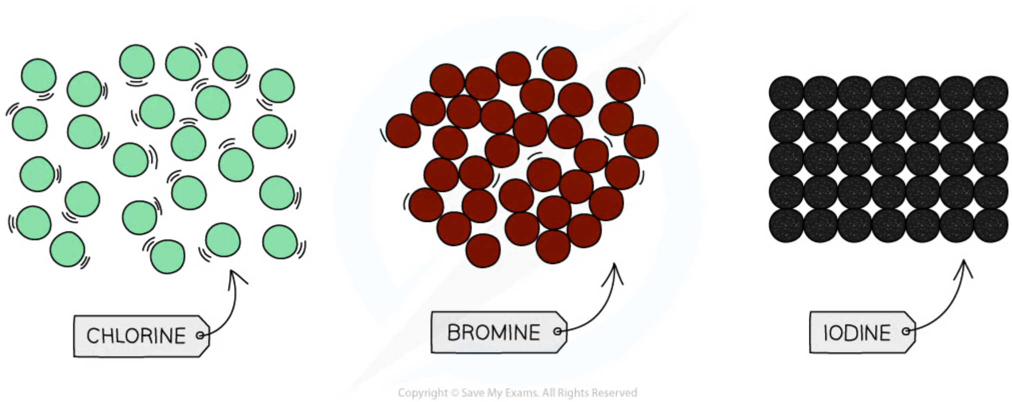
Group VII Properties & Trends

The halogens

- These are the Group VII non-metals that are **poisonous** and include fluorine, chlorine, bromine, iodine and astatine
- Halogens are **diatomic**, meaning they form molecules of **two** atoms
- All halogens have seven electrons in their outer shell
- They form halide ions by gaining one more electron to complete their outer shells
- Fluorine is not allowed in schools so observations and experiments tend to only involve chlorine, bromine and iodine

Properties of the halogens

- At room temperature (20 °C), the physical state of the halogens changes as you go down the group
- Chlorine is a **pale yellow-green gas**, bromine is a **red-brown liquid** and iodine is a **grey-black solid**
- This demonstrates that the **density** of the halogens **increases** as you go **down the group:**



The physical state of the halogens at room temperature

- **Reactivity** of Group VII non-metals **increases** as you go **up** the group (this is the opposite trend to that of Group I)
- Each outer shell contains seven electrons and when the halogen reacts, it will need to gain one outer electron to get a full outer shell of electrons
- As you go up Group VII, the number of shells of electrons **decreases** (period number decreases moving up the Periodic Table)
- This means that the outer electrons are **closer** to the nucleus so there are **stronger** electrostatic forces of attraction, which help to attract the extra electron needed
- This allows an electron to be attracted more readily, so the **higher** up the element is in Group VII then the **more reactive** it is

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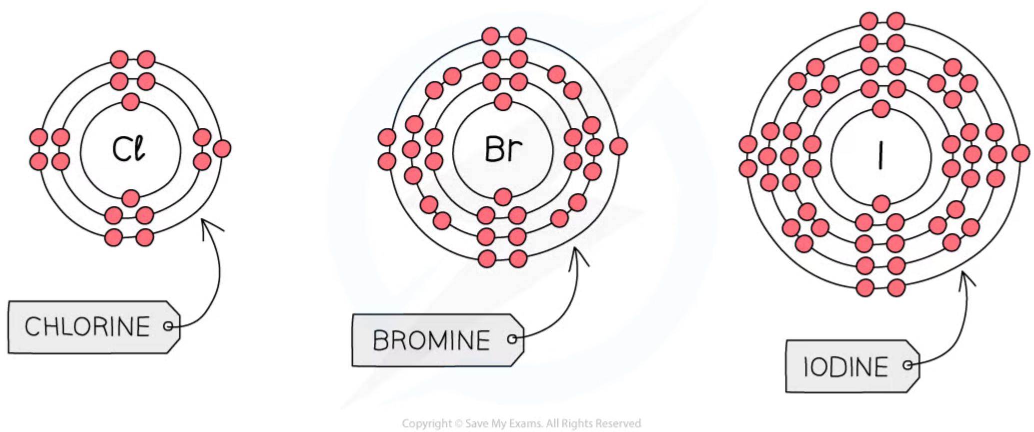

Diagram showing the electronic configuration of the first three elements in Group VII

Exam Tip

Solid iodine, iodine in solution and iodine vapour are different colours. Solid iodine is dark **grey-black**, iodine vapour is **purple** and aqueous iodine is **brown**.

Predicting Group VII Properties

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- You may be given information about some elements and asked to predict the properties of other elements in the group
- The information you might be given could be in relation to melting/boiling point or physical state/density so it is useful to know the trends in properties going down the group

Melting and boiling point

- The melting and boiling point of the halogens **increases** as you go down the group
- Fluorine is at the top of Group VII so will have the **lowest** melting and boiling point
- Astatine is at the bottom of Group VII so will have the **highest** melting and boiling point

Physical states

- The halogens become **denser** as you go down the group
- Fluorine is at the top of Group VII so will be a **gas**
- Astatine is at the bottom of Group VII so will be a **solid**

Colour

- The colour of the halogens becomes **darker** as you go down the group
- Fluorine is at the top of Group VII so the colour will be **lighter**, so fluorine is **yellow**
- Astatine is at the bottom of Group VII so the colour will be **darker**, so astatine is **black**

Exam Tip

If you are doing the supplement / extended course you can be asked to identify trends in chemical or physical properties of the Group VII elements, given appropriate data.

Firstly, make sure that you have placed the elements and associated data in either ascending or descending order according to their position in Group VII. Then look for any general patterns in the data.

8.2.3 Group VII Displacement Reactions

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Group VII Displacement Reactions

- A halogen displacement reaction occurs when a more reactive halogen displaces a less reactive halogen from an aqueous solution of its halide
- The reactivity of Group VII non-metals increases as you move up the group
- Out of the three commonly used halogens, chlorine, bromine and iodine, chlorine is the most reactive and iodine is the least reactive

Colour of Halogens in Aqueous Solutions

AQUEOUS SOLUTION	COLOUR
CHLORINE	VERY PALE GREEN, BUT USUALLY APPEARS COLOURLESS AS IT IS VERY DILUTE
BROMINE	ORANGE BUT WILL TURN YELLOW WHEN DILUTED
IODINE	BROWN

Halogen displacement reactions

Chlorine and bromine

- If you add chlorine solution to colourless potassium bromide solution, the solution becomes orange as bromine is formed
- Chlorine is **above** bromine in Group VII so is more reactive
- Chlorine will therefore **displace** bromine from an aqueous solution of the metal bromide
- The least reactive halogen always ends up in the elemental form

potassium bromide + chlorine → potassium chloride + bromine

$$2KBr\ (aq) + Cl_2\ (aq) \rightarrow 2KCl\ (aq) + Br_2\ (aq)$$

Bromine and iodine

- Bromine is above iodine in Group VII so is **more** reactive
- Bromine will therefore displace iodine from an aqueous solution of metal iodide
- The solution will turn brown as iodine is formed

magnesium iodide + bromine → magnesium bromide + iodine

$$2MgI\ (aq) + Br_2\ (aq) \rightarrow 2MgBr\ (aq) + I_2\ (aq)$$

Summary table of displacement reactions

	Chlorine (Cl_2)	Bromine (Br_2)	Iodine (I_2)
Potassium chloride (KCl)	X	No reaction	No reaction
Potassium bromide (KBr)	Chlorine displaces the bromide ions. Yellow–orange colour of bromine is seen	X	No reaction
Potassium iodide (KI)	Chlorine displaces the iodide ions. Brown colour of iodine is seen	Bromine displaces the iodide ions:. Brown colour of iodine is seen	X

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Exam Tip

Iodine solid, solution and vapour are different colours. Solid iodine is dark **grey-black**, iodine vapour is **purple** and aqueous iodine is **brown**.

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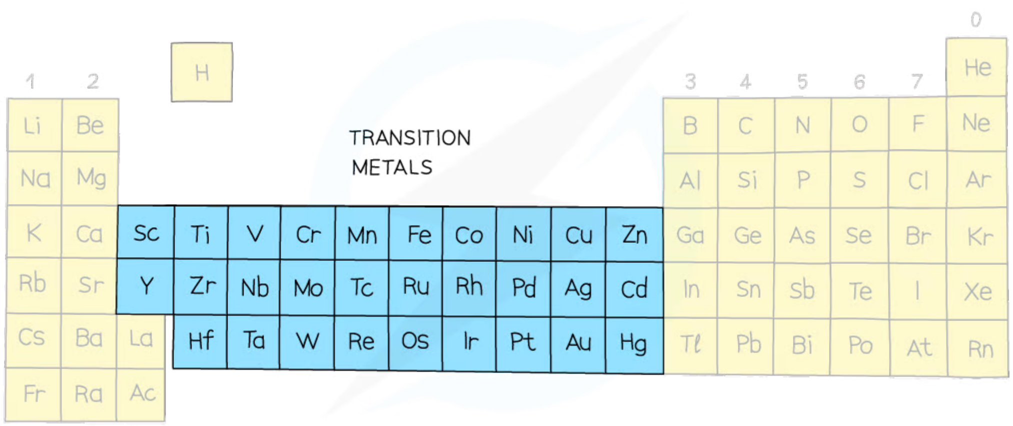
8.2.4 Transition Elements

Transition Elements

General properties of the transition elements

- They are very **hard** and strong metals and are good conductors of **heat** and **electricity**
- They have very **high melting** points and are highly **dense** metals
- For example, the melting point of titanium is 1,688°C whereas potassium in Group I melts at only 63.5°C, slightly warmer than the average cup of hot chocolate!
- The transition elements form **coloured** compounds and often have more than one oxidation state, such as iron readily forming compounds of both Fe^{2+} and Fe^{3+}
- These coloured compounds are responsible for the pigments in many paints and the colours of gemstones and rocks
- Transition elements, as elements or in compounds, are often used as **catalysts** to improve the rate or reaction in industrial processes
    - Transition element catalysts of platinum or rhodium are also used in car exhausts in the 'catalytic convertor' to reduce the levels of nitrous oxides and carbon monoxide produced



The transition elements on the Periodic Table

Exam Tip

Although scandium and zinc are in the transition element area of the Periodic Table, they are not considered transition elements as they do **not** form coloured compounds and have only **one** oxidation state.

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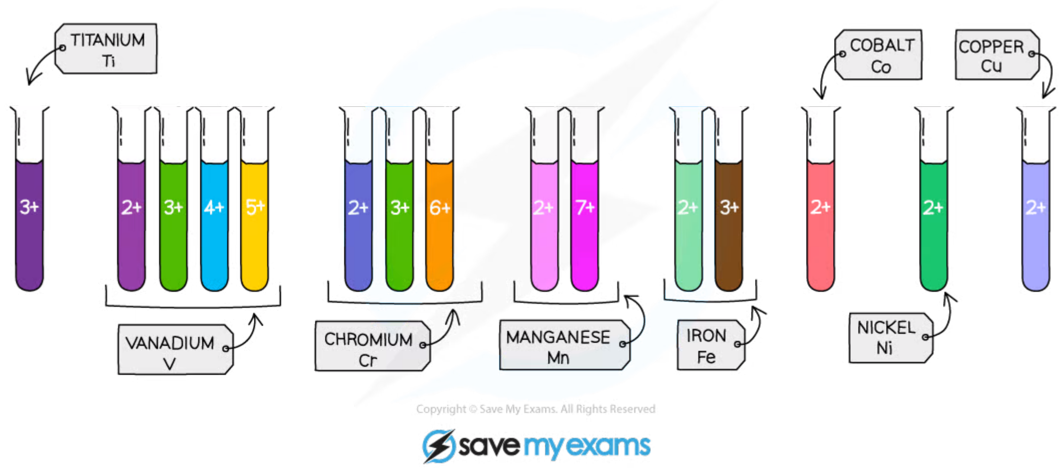
Transition Elements Oxidation Numbers

EXTENDED

- The transition elements have more than one **oxidation number**, as they can lose a different number of electrons, depending on the chemical environment they are in
- For example. iron either:
  - Lose **two** electrons to form Fe^{2+} so has an oxidation number of +2
  - Loses **three** electrons to form Fe^{3+} so has an oxidation number of +3
- Compounds containing transition elements in different oxidation states will have different **properties** and colours



Ions of the same element can have different oxidation numbers forming different colours

Uses of the transition elements

- The transition elements are used extensively as **catalysts** due to their ability to interchange between a range of oxidation states
- This allows them to form complexes with reagents which can easily **donate** and **accept** electrons from other chemical species within a reaction system
- They are used in **medicine** and **surgical** applications such as limb and joint replacement (titanium is often used for this as it can bond with bones due to its high biocompatibility)
- They are also used to form coloured compounds in **dyes** and **paints**, **stained** glass **jewellery**

8.2.5 Noble Gases

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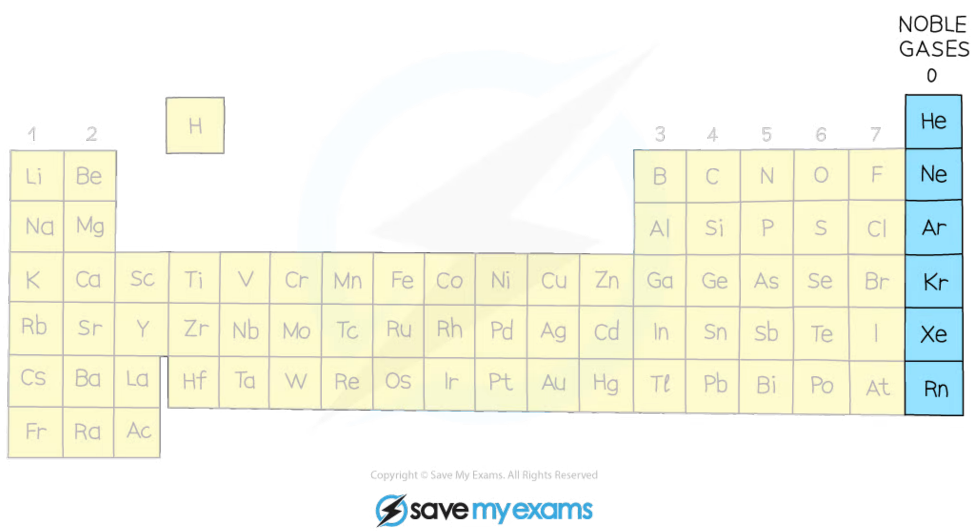
Noble Gases Properties & Electronic Configuration

The Noble Gases

- The noble gases are in Group VIII (or Group 0); they are non-metals and have very **low melting** and **boiling** points
- They are all **monoatomic**, **colourless** gases
- The Group 0 elements all have **full outer shells**
- This electronic configuration is **extremely stable** so these elements are unreactive and are inert
- Electronic configurations of the noble gases:
  - He: 2
  - Ne: 2,8
  - Ar: 2,8,8
  - Kr: 2,8,18,8
  - Xe: 2,8,18,18,8



Noble gases are inert (unreactive) as they have a full outer shell of electrons so do not easily lose or gain electrons